INTEL PRECISE TOUCH ENABLEMENT GUIDE FOR LNL-MX ELAN BOM 36A/37A – UPDATE WW 27' 2023

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LNL- MX

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J3C2 Touch Conn 1

J2C1 Touch Conn 2

BOM 36A/37A -HID SPI CABLE

Connected to Panel

Connected to LNL Mx RVP

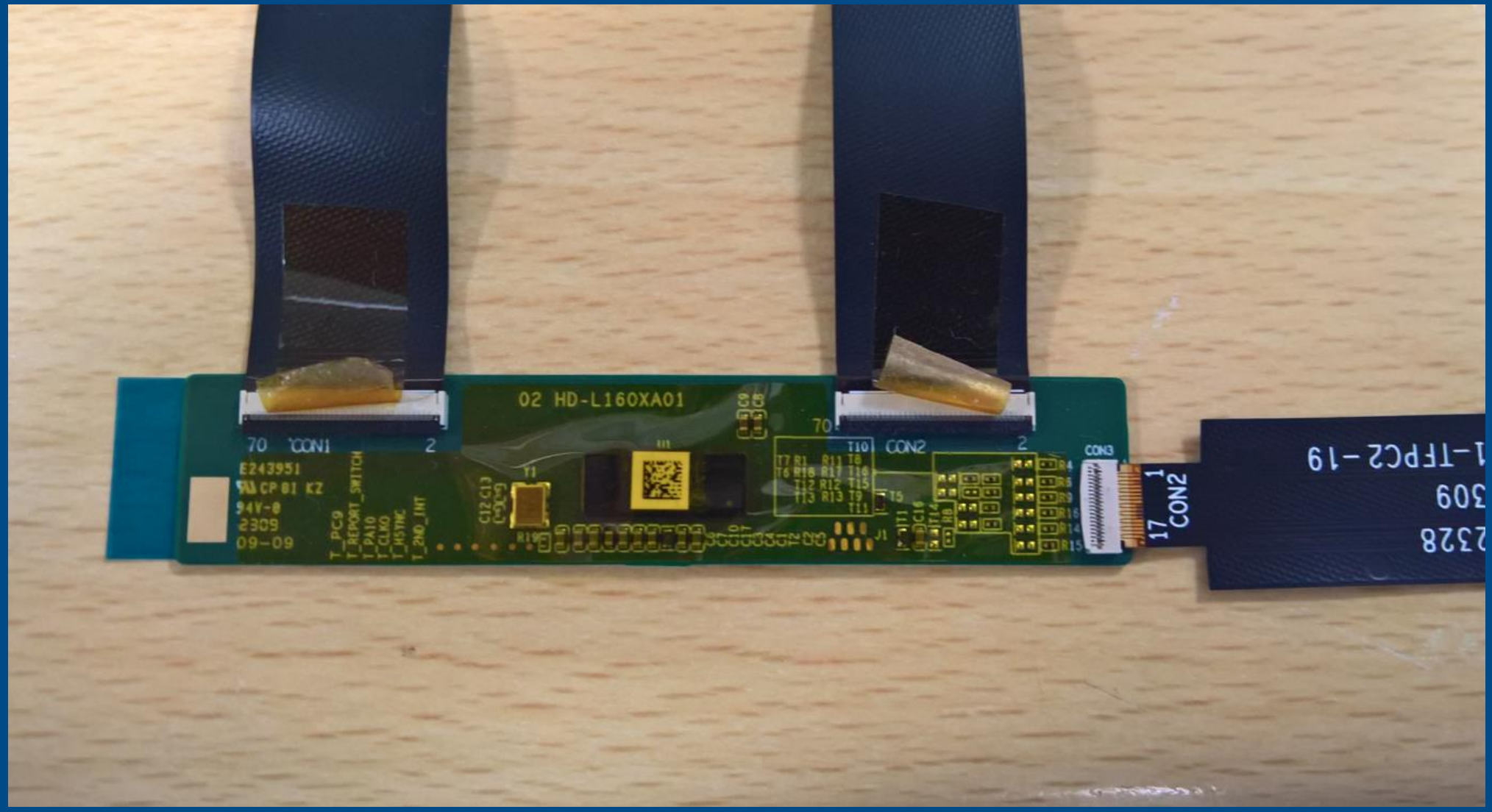
PART - II

ELAN BOM 36A/37A PANEL

ELAN BOM 36A/37A PANEL

ELAN CONTROLLER (BACK)

ELAN CONNECTOR



LNL- MX RVP SPI REWORK & INTERFACE (1/2)

Connect Touch Controller to below ports through SPI cable:

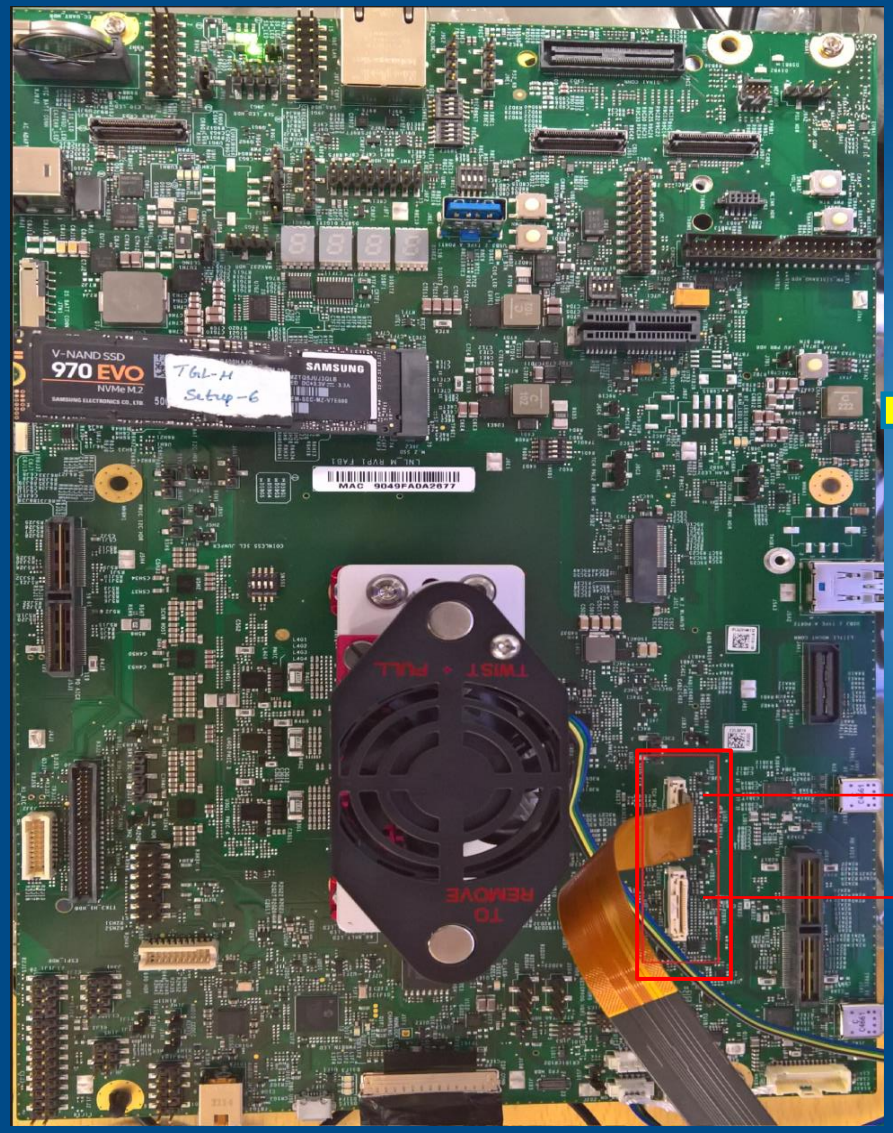
- J3C2 THC 0 (Port 1 in Windows)
- J2C1 THC 1 (Port 2 in Windows)

Rework

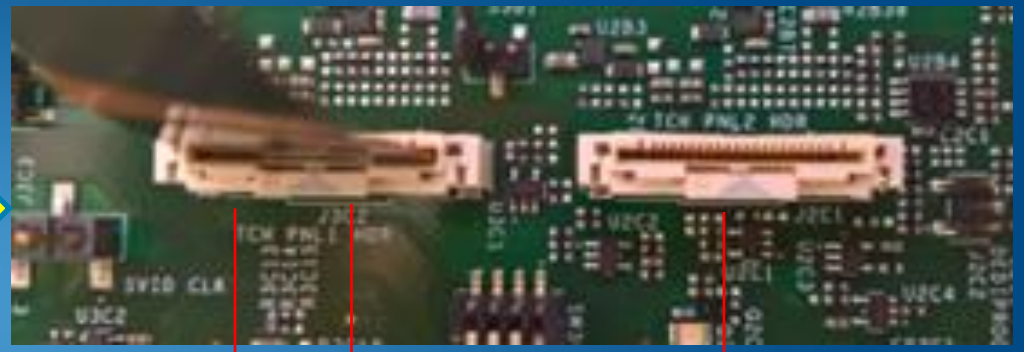
- R01 Rework - Document to avoid CMOS RTC battery self-drain when it is left idle for long time
- F-19 Rework – To enable SPI on Port 2.

RVP SPI INTERFACE (2/2)

LNL Mx RVP



Zoomed Version of Touch Connectors



THC0(Port1) Connector1 → SPI(Default)

THC1(Port2) Connector2 → I2C

J3C2 Touch Conn 1

J2C1 Touch Conn 2

Push pin to remove SPI/I2C Cable from Port

PART - III
BIOS SETTINGS

BIOS SETTINGS FOR THC BASE DRIVER

- Navigate to BIOS Menu by pressing F2 during System Boot.
- Navigate to : Intel Advanced Menu → PCH-IO Configuration → THC Configuration and enable the below mentioned configuration:
  - ✓ THC Port Configuration = THC0[Mandatory for both port to be functional]
  - ✓ Port Clock = Functional (Default)
  - ✓ THC Port Configuration = THC1
  - ✓ Port Clock = Functional (Default)
  - ✓ For enabling THC1 the below setting is mandatory.
  - ✓Intel Advanced Menu → PCH-IO Configuration → Serial IO Configuration → Disable I2C 0 to I2C 5 and SPI 0 to SPI 2.

BIOS SETTINGS FOR QUICK SPI DRIVER

BIOS Settings FOR THC 0 (Port 1)

- Navigate to BIOS Menu by pressing F2 during System Boot.
- Navigate to: Intel Advanced Menu → PCH-IO Configuration → THC Configuration and enable below mentioned configurations :

✓ Intel Advanced Menu→ PCHIO Configuration→THC Configuration → THC Port Configuration → THC0

✓ Intel Advanced Menu→ PCHIO Configuration→THC Configuration → HID Over SPI Limit Packet Size → 0

✓ Intel Advanced Menu→ PCHIO Configuration→THC Configuration → THC Port Configuration → THC Mode → HID OVER SPI.

✓ Intel Advanced Menu→ PCHIO Configuration→ THC Configuration → THC Port Configuration →Input Report Body Address→ **[1100]**

✓ Intel Advanced Menu→ PCHIO Configuration→THC Configuration → THC Port Configuration → Input Report Header Address→ [1000]

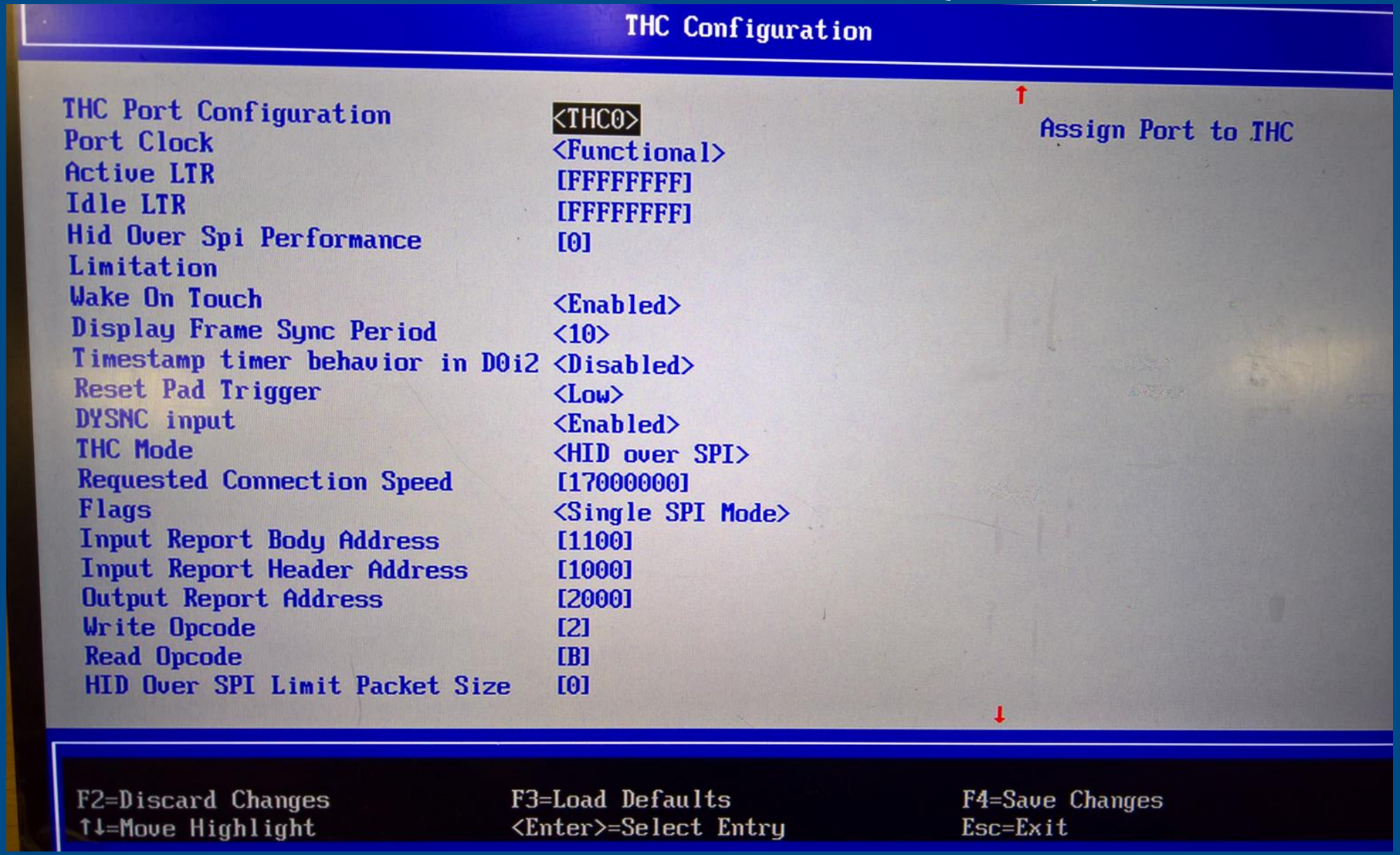
✓ Intel Advanced Menu→ PCHIO Configuration→THC Configuration → THC Port Configuration → Output Report Address→ [2000]

✓ Intel Advanced Menu→ PCHIO Configuration→THC Configuration → THC Port Configuration → Write Opcode→ [2].

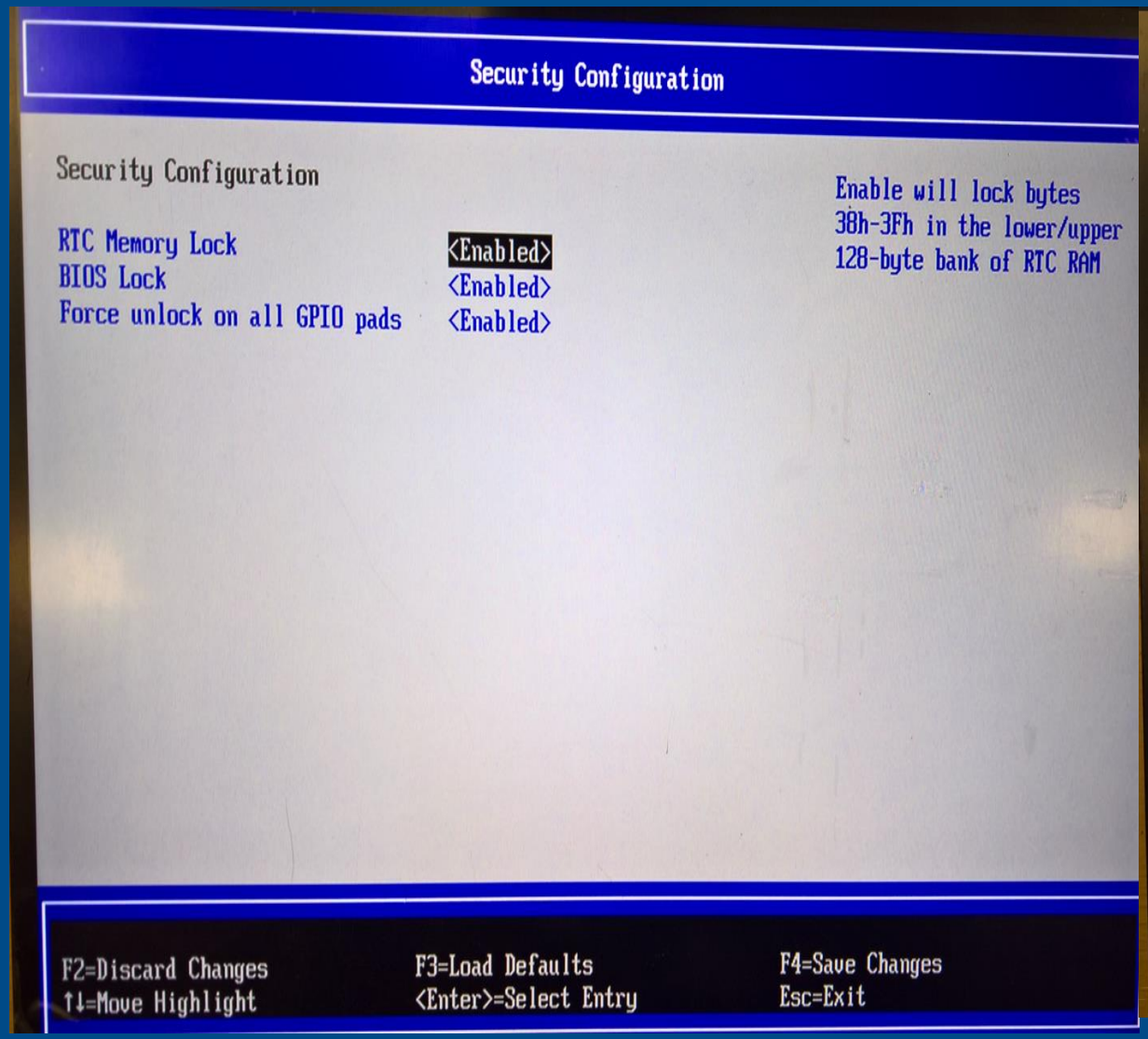
✓ Intel Advanced Menu→ PCHIO Configuration→ THC Configuration → THC Port Configuration → Read Opcode→ [B].

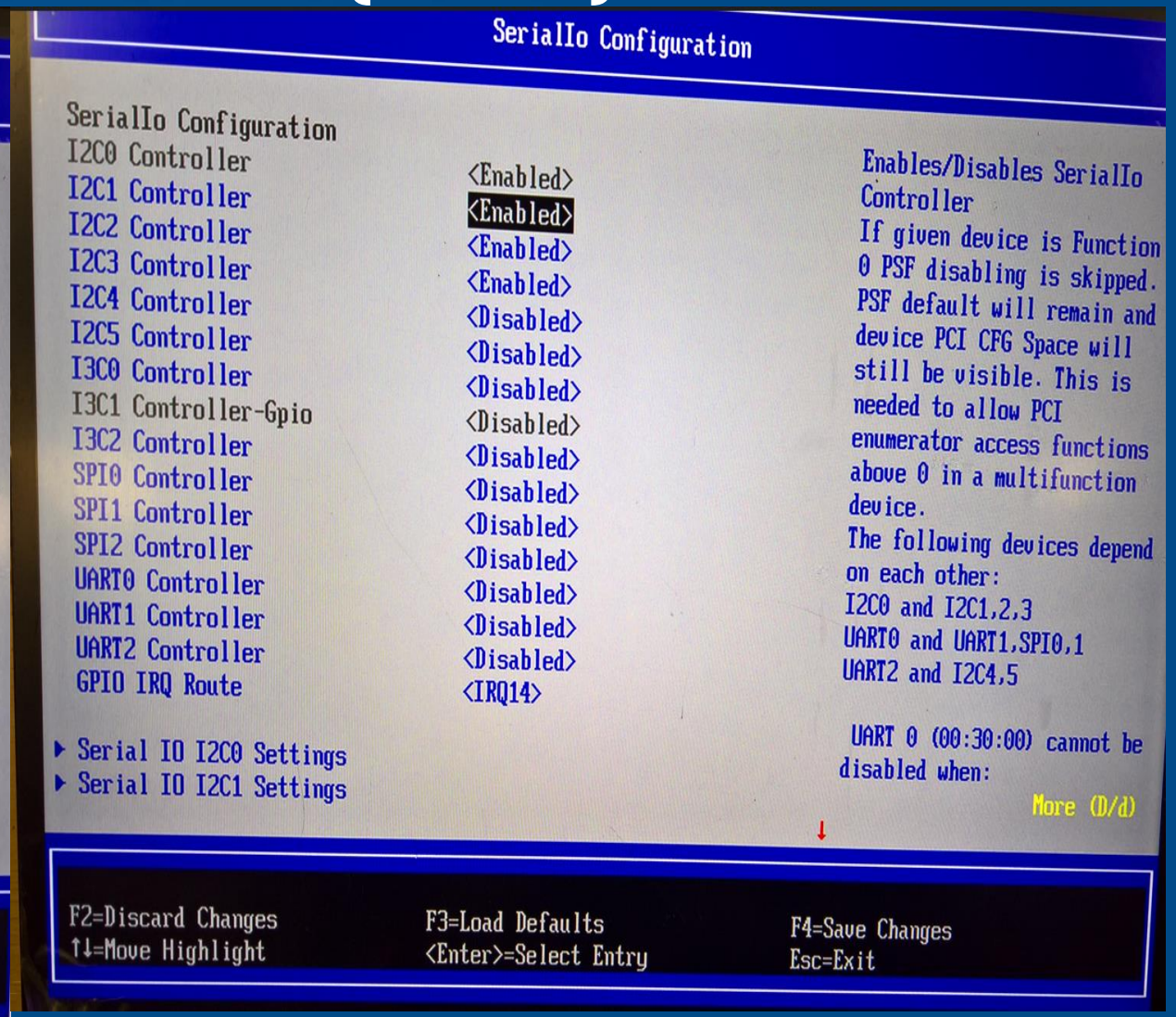
✓ **Intel Advanced Menu → PCH-IO Configuration → Serial IO Configuration → Disable I2C 0 to I2C 5 and SPI 0 to SPI 2.**

BIOS SETTINGS FOR THC 0 (PORT 1)



BIOS SETTINGS FOR THC 0 (PORT 1)





INSTALLATION OF INTEL QUICK SPI DRIVER

- Verified on Windows11 22h2 (22621.X) .
- There are a few ways to install driver, below are the methods:
  - Install by right “.inf” file and select “install” .
  - Upgrade driver through Device Manager.
  - Install driver through PNP CMD.

    CMD: pnputil /add-driver IntelQuickSPI.inf /install

END